Payload Estimation

By Aryan Verma

Theoretical Modelling

Key environmental, electrical, and aerodynamic parameters influencing payload estimation.

Propulsion & Power

- **Motor Thermal Efficiency**
  Impacts power consumption as temperatures rise.
- **Battery State of Charge**
  Reduced battery voltage increases the current consumption which then increases the heat generated.
- **Motor Aging**
  Gradually degrades motor performance and increases power draw.

Environmental

- **Air Density**
  Affects lift generation and aerodynamic drag on the chassis.
- **Wind**
  Introduces dynamic forces requiring continuous stability correction and increasing power draw

Aerodynamics

- **Thrust Blockage**
  Airflow restrictions from structural frames.
- **Ground Effect**
  Alters aerodynamic lift forces during low-altitude hover.

$$P_{total} = k_b * w + k\,w^n + I^2 R + k_w\, w^3 + \frac{(T_{total})^{3/2} * (1/\eta)}{\sqrt{2 * \rho * PropArea}}$$

$k_b * w$ → Bearing Loss

$k\,w^n$ → Hysteresis Losses

$I^2 R$ → Resistance Loss

$k_w\, w^3$ → Propeller Drag Loss

$\frac{(T_{total})^{3/2} * (1/\eta)}{\sqrt{2 * \rho * PropArea}}$ → Actual Power Consumption

$$\boxed{T_{total}^2 = D\_wind^2 + T_{actual}^2}$$

$$R = R_0 + \alpha[T - T_0]$$

$\boxed{\alpha = 0.00393}$ $T_0 = 25°C$ or $298K$

$\cancel{R_0 = 0}$

A_lost = 0.0081 m^2 (2.7cm * 27cm) (Thrust Blockage)

Ground effect → Presented Later

Voltage (V)	Propeller	Throttle (%)	Thrust (g)	Ampere (A)	Power (W)	Speed (RPM)	Efficiency (g/W)
46V (12S LIPO)	HW 24*8.0 Inch Foldable Propeller	40%	2416	4.6	212.2	2455	11.3
		42%	2732	5.5	253.5	2605	10.6
		44%	3058	6.4	296.6	2754	10.3
		46%	3377	7.4	343.8	2900	9.7
		48%	3746	8.7	401.8	3047	9.2
		50%	4106	9.9	460.2	3184	8.9
		52%	4419	11.2	519.9	3325	8.4
		54%	4822	12.5	580.2	3456	8.3
		56%	5209	14.1	653.4	3589	8.0
		58%	5476	15.6	724.6	3716	7.6
		60%	5947	17.3	800.2	3841	7.4
		62%	6370	19.1	886.2	3959	7.2
		64%	6709	20.7	958.4	4080	7.0
		66%	7086	22.6	1048.2	4194	6.7
		68%	7501	24.9	1153.8	4307	6.5
		70%	7779	26.6	1230.2	4415	6.3
		72%	8238	28.8	1332.1	4519	6.2
		74%	8654	31.0	1435.8	4627	6.0
		76%	9016	32.7	1514.0	4723	5.9
		78%	9294	35.1	1626.5	4821	5.7
		80%	9782	37.6	1742.3	4912	5.6
		90%	10731	43.6	2005.6	5142	5.3
		100%	11822	51.8	2399	5442	4.9

Motor Efficiency

```
# Temp = 20 # in degrees
# delta_T = Temp - 20 # in degrees
# alpha_copper = 0.00393  # per degree C

def RHS(X, R_copper, k, n, k_b, k_w, prop_coeff):
    current_A, rps, thrust_N = X  # unpack your two variables
    return R_copper * current_A **2 + k * rps**n + k_b * rps + k_w * rps**3 + (1/prop_coeff) * (thrust_N ** 1.5 ) / denom

popt, pcov = curve_fit(RHS, (current_A, speed_rps, thrust_N), LHS, p0=(0.34, 1e-6, 1.5, 1e-7, 1e-9, 0.7), bounds = ([0,0, 1, 0, 0,
0.65], [1, 1e-2, 2, 1e-3, 1e-3, 0.95]))
R_copper, k, n, k_b, k_w, prop_coeff = popt


print(f"R_copper: {R_copper:.5f} ohms")
print(f"k: {k:.6e}")
print(f"n: {n:.4f}")
print(f"k_b: {k_b:.6e}")
print(f"k_w: {k_w:.6e}")
print(f"prop_coeff: {prop_coeff:.6e}")
```

```
R_copper: 0.09444 ohms
k: 1.700603e-03
n: 1.6832
k_b: 1.535252e-07
k_w: 2.755471e-06
prop_coeff: 9.499974e-01
```

Air Density

$$P_{total} = \underbrace{k_b * w}_{\text{Bearing Loss}} + \underbrace{k_c w^n}_{\text{Hysteresis Losses}} + \underbrace{I^2 R}_{\text{Resistance Loss}} + \underbrace{k_w w^3}_{\text{Propeller Drag Loss}} + \underbrace{\frac{(T_{total})^{3/2} * (1/\eta)}{\sqrt{2 * \rho * \text{Prop Area}}}}_{\text{Actual Power Consumption}}$$

$$\boxed{T_{total}^2 = D\_wind^2 + T_{actual}^2}$$

$$R = R_0 + \alpha[T - T_0]$$

$\boxed{\alpha = 0.00393}$ $T_0 = 25°C$ or $298K$

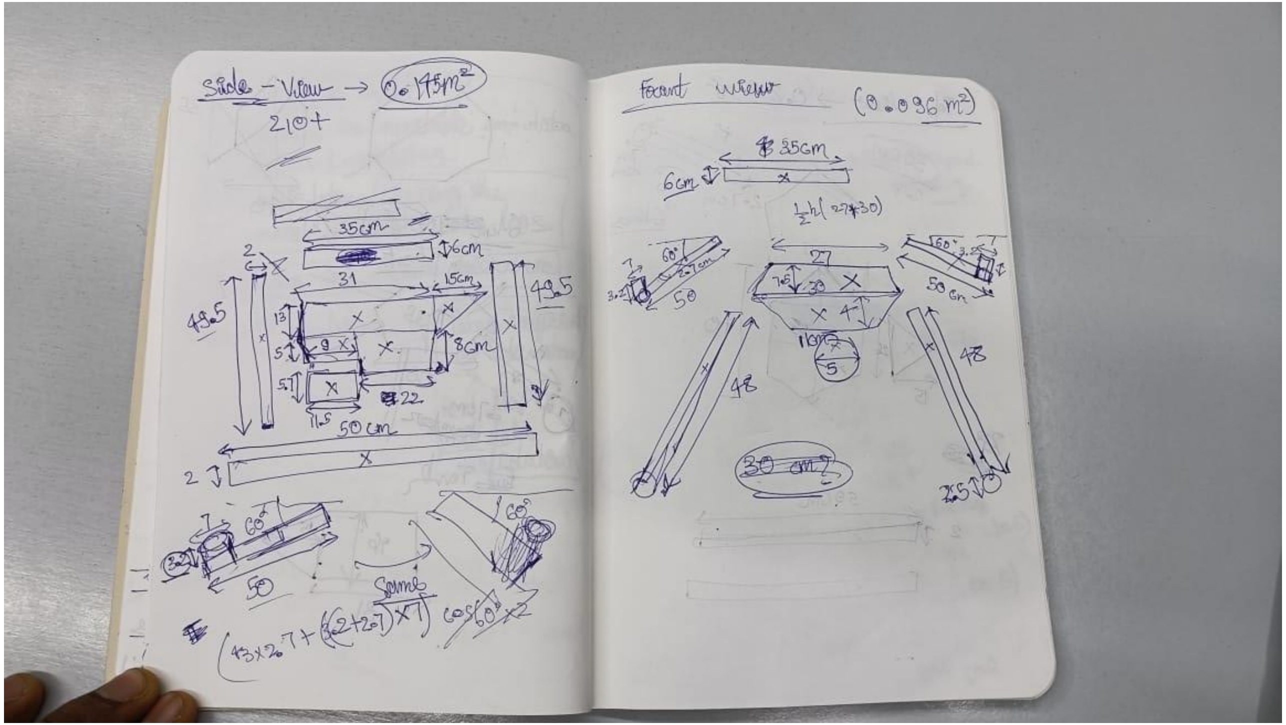
$\cancel{R_0 = 0}$

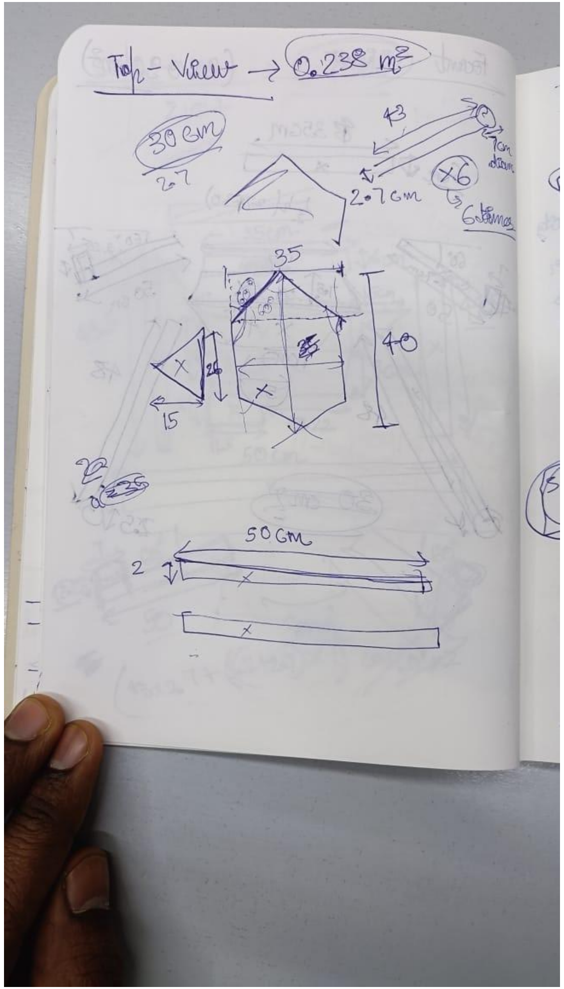
$A\_lost = 0.0081\ m^2$ $(2.7cm * 27cm)$ (Thrust Blockage)

Ground effect → Presented Later

Wind = 0.5 *density * v_wind^2 * A * C_d

C_d = 0.4 +- 0.1 (From literary review on DJI Agras class drones)





Thrust Blockage

Ground effect correction at 2m AGL is 0.22%

$$\frac{P_{IGE}}{P_{OGE}} = \left[1 - \left(\frac{R}{4Z}\right)^2\right]^{3/2}$$

R → Rotor Diameter → 24 inches

Z → Distance from Ground